Enhancing IoMT Security Using Large Multimodal Models

Yan Jiao, Mian Muaz Razaq, Limei Peng, and Pin-Han Ho

Abstract

This paper explores the potential of Large Multimodal Models (LMMs) in enhancing the security of Internet of Medical Things (IoMT) networks. As IoMT systems become more complex and generate vast amounts of sensitive data, traditional security measures are increasingly insufficient to address evolving cyber threats. The ability of LMMs to process and integrate diverse data types, such as sensor data, audio, and video, offers a powerful approach to detecting and mitigating security risks in real time. By focusing on key applications like anomaly detection through traffic prediction and improving Physical Layer Security (PLS), this work demonstrates how LMMs can significantly enhance the resilience of IoMT networks against a wide range of cyber threats. The integration of multimodal data enables more accurate traffic predictions and allows adaptive responses to emerging security challenges. Additionally, LMMs contribute to strengthening PLS by identifying vulnerabilities in communication channels and optimizing resource allocation. Through dynamic data fusion techniques, LMMs ensure that IoMT systems can respond to threats in real-time, while maintaining high security and minimizing impact on system performance.

Introduction

The Internet of Medical Things (IoMT) is transforming healthcare by interconnecting medical devices, sensors, and systems, generating vast amounts of diverse, highly sensitive data. This connectivity reshapes how medical information is collected, processed, and utilized, spanning applications from wearable health monitors to advanced diagnostic tools. These advancements significantly enhance patient care through continuous monitoring, personalized treatment plans, and improved clinical outcomes. However, the increasing interconnectivity of IoMT devices introduces substantial security challenges. The complexity and sensitivity of IoMT data necessitate robust security measures to protect patient information from breaches, unauthorized access, and other cyber threats—concerns that are becoming increasingly urgent as IoMT networks expand and evolve.

In addition to general IoT security schemes, such as encryption, device authentication, network segmentation, and anomaly detection using machine learning (ML) and rule-based systems, IoMT healthcare organizations employ private networks and prepare redundant backup devices to enhance network security and robustness. These measures safeguard data flow, ensure device integrity, detect unusual activities, and enable prompt responses to failures or attacks that could indicate security threats [1]. Nevertheless, the exponential growth of medical data and its increasing diversity have added significant complexity to communication systems and data integration management. Recent advancements in diversity-aware frameworks, such as those proposed in proactive caching strategies for mobile networks, demonstrate the importance of optimizing data distribution across heterogeneous systems to meet scalability and performance demands [2]. Consequently, conventional methods often fail to meet the escalating security and safety requirements of IoMT organizations. This challenge is critical, as the security and confidentiality of medical devices are directly tied to patient health and safety, with far-reaching implications. Achieving satisfactory network security and information confidentiality demands real-time and efficient data management.

Anomaly detection is particularly vital in IoMT, as it enables the early identification of attacks or failures, providing administrators with the opportunity to address potential security issues proactively and minimize the risk of major medical incidents. Physical Layer Security (PLS) is equally important for maintaining a robust and secure IoMT framework. Considering the unique characteristics of IoMT applications, such as diverse data formats (text, image, video), varied signal types (e.g., RF, wired/wireless, frequency bands, multiplexing technologies), and the high volume of health-relevant data, Large Multimodal Models (LMMs) offer a promising approach. LMMs can process, integrate, and analyze multimodal IoMT data. By employing traffic prediction based on fused multimodal data, LMMs enhance anomaly detection and PLS by leveraging the complementary strengths of diverse data types, offering a more comprehensive view of security risks than single-modality models. This approach enhances anomaly detection accuracy, reduces false positives, and improves contextual awareness, enabling adaptive threat responses while

Digital Object Identifier: 10.1109/MNET.2025.3545692
Date of Current Version: 15 July 2025
Date of Publication: 25 February 2025

Yan Jiao and Pin-Han Ho (corresponding author) are with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518000, China; Mian Muaz Razaq is with the School of Computer Science and Engineering, Kyungpook National University, Daegu 41566, South Korea; Limei Peng (corresponding author) is with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518000, China, and also with the School of Computer Science and Engineering, Kyungpook National University, Daegu 41566, South Korea.

0890-8044 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies.

Authorized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

strengthening security in complex IoMT environments. Nonetheless, the application of LMMs in IoMT for proactive anomaly detection to prevent network anomalies and enhance communication security is still in its infancy and faces a series of unprecedented challenges requiring urgent exploration and solutions.

In this paper, we propose leveraging LMMs to enhance IoMT network security through two primary objectives: traffic prediction for anomaly detection and improved Physical Layer Security (PLS), as shown in Fig. 1. Traffic prediction focuses on utilizing LMMs to analyze and integrate multimodal IoMT data and signals, such as text, image, video, and signals from base stations and Wi-Fi networks, to accurately predict network traffic patterns. This capability enables the early detection of anomalies indicative of security threats, such as potential attacks or system failures. For PLS, we explore the application of LMMs to strengthen the IoMT framework by identifying vulnerabilities, optimizing resource allocation, and providing robust defense mechanisms at the physical layer.

The rest of this paper is organized as follows. The section "State-of-the-Art LLM/Multimodality-Based Methods in Network Management" introduces the state-of-the-art work related to LMMs. The sections "LMM-Based Anomaly Detection" and "LMM-Based Data Fusion for Physical Layer Security (PLS) in IoMT" present the framework, challenges, and solutions for LMM-based anomaly detection and PLS, respectively. The section "Case Study" provides a case study. The section "Conclusion" concludes the paper.

State-of-the-Art LLM/Multimodality-Based Methods in Network Management

Addressing IoMT security challenges, such as anomaly detection and Physical Layer Security (PLS), requires leveraging LMMs alongside insights from state-of-the-art methods. This section reviews recent Large Language Model (LLM)- and multimodality-based techniques, focusing on their principles, limitations, and relevance to traffic prediction, log analysis, and resource allocation, as summarized in Table 1.

Traffic Prediction for IoMT

Traffic prediction is vital for IoMT to enable anomaly detection and ensure efficient resource management in highly dynamic environments. Current research has explored LLM-based approaches, such as a framework using in-context learning for wireless traffic prediction [3]. While effective in avoiding fine-tuning, these methods often rely solely on textual data, underutilizing the potential of multimodal information inherent in IoMT, such as signal patterns and device-specific behaviors. Other studies employ multimodal deep learning frameworks [4] or meta-learning techniques [5], but these are constrained by short-term predictions, reliance on simulated data, or assumptions of uniform data distributions. Addressing these gaps requires LMM-based approaches capable of fusing diverse IoMT data modalities for accurate, multi-step predictions in real-world healthcare scenarios.

Log Analysis in IoMT

Log analysis is essential for IoMT network monitoring, anomaly detection, and fault diagnosis. Existing LLM-based methods [6], [7] have demonstrated promise in parsing and analyzing network logs but often depend on labor-intensive manual annotations or fail to consider attributes like timestamps and log security levels critical to IoMT environments. Furthermore, root cause analysis systems [8] lack adaptability to unmonitored incidents or predefined categories, and fine-tuned models face challenges in acquiring consistent reliability. To address these shortcomings, IoMT log analysis requires advanced LMMs that can integrate multimodal log attributes and historical data, providing explainable, real-time insights for proactive anomaly detection and resolution.

Resource Allocation in IoMT

Resource allocation in IoMT ensures efficient use of network resources, critical for maintaining uninterrupted healthcare services. Recent

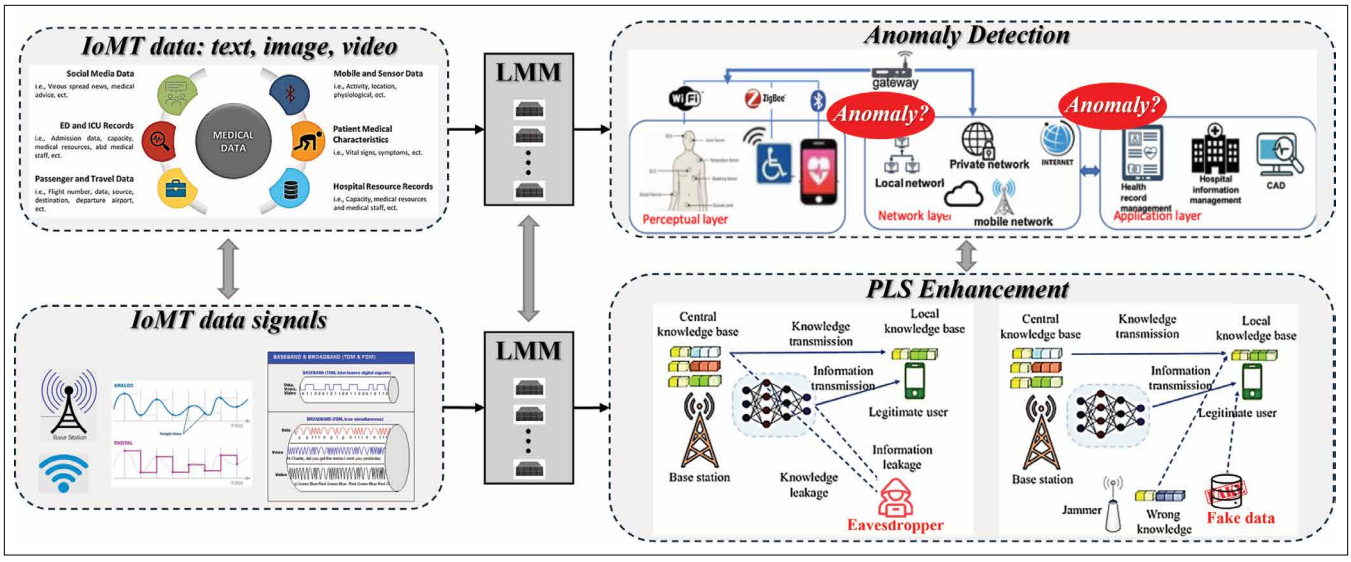


FIGURE 1. Framework of the LMM-based Anomaly Detection and PLS enhancement in IoMT.

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

LLM-based methods [9], [10], [11] show potential in wireless communication, network slicing, and virtual network functions (VNF). However, these approaches often overlook the multimodal nature of IoMT traffic, including real-time bandwidth and memory demands, and are constrained by computational inefficiencies and transparency issues. Leveraging LMMs for resource allocation can address these limitations by incorporating real-time multimodal data, improving the adaptability and efficiency of resource management in IoMT networks.

Challenges and Opportunities for LMMs in IoMT

While research predominantly focuses on LLMs or small multimodal models, applying LMMs to IoMT remains largely unexplored. Challenges include integrating diverse IoMT data modalities, achieving real-time processing with high computational efficiency, and ensuring explainability in decision-making processes. However, the unique capabilities of LMMs to fuse and analyze multimodal data present immense potential for advancing IoMT traffic prediction, log analysis, and resource allocation, ultimately enhancing network security and operational resilience.

LMM-Based Anomaly Detection

Proposed Framework

This section introduces the use of LMM-based traffic prediction for anomaly detection in IoMT, as illustrated in Fig. 2. By leveraging the multimodal capabilities of LMMs, the framework processes diverse data types, such as sensor readings, network logs, and audio-visual inputs, to predict traffic patterns and identify potential anomalies.

The process begins with *data preprocessing*, where raw multimodal IoT data is standardized and transformed to address heterogeneity and high dimensionality. Techniques such as tokenization for text data, frame sampling for video data, spectrogram generation for audio data, and noise reduction for time-series data ensure that meaningful features are extracted while preserving temporal and spatial characteristics. Next, these preprocessed features are integrated into the LMM using advanced *fusion strategies*, such as early fusion, intermediate fusion, or late fusion. Early fusion combines raw data from different modalities at the input level, enabling the model to learn joint representations from the outset. Intermediate fusion integrates feature-level representations,

Tasks	References	Model	Data type	Goal
Traffic prediction	TrafficLLM [3]	GPT-4	Unimodal, time-seires	Wireless traffic load volume prediction
	CNN-GNN [4]	ConvLSTM, AGCN	Multimodal, grid + graph	Mobile traffic matrix prediction
	MetaSTNet [5]	Encoder-decoder, GCN, CNN	Multimodal, time-series + text+image	Cellular traffic prediction
Log analysis	ONLA-LLM [6]	LLaMA2	Unimodal, optical network logs	Log parsing, anomaly detection and classification, anomaly analysis, report generation
	LogBERT [7]	ChatGPT-3.5, BERT	Unimodal, network logs	Log parser and log anomaly detection
	RCACopilot [8]	GPT-3.5-turbo, GPT-4	Unimodal, cloud incidents	Automated root cause analysis
Resource allocation	[9]	LLaMA	Unimodal, channel gains-transmit power strategies	Allocate transmit power in wireless communication
	WirelessAgent [10]	GPT-4o-128K	Multimodal, sensor readings, 2D/3D vision and radio signals	Wireless network slicing management
	[11]	LLaMA2	Unimodal, historical VNF resource consumption	Forecast VNF resource consumption

TABLE 1. A summary of works on network management tasks using LLM/Multimodality-based methods.

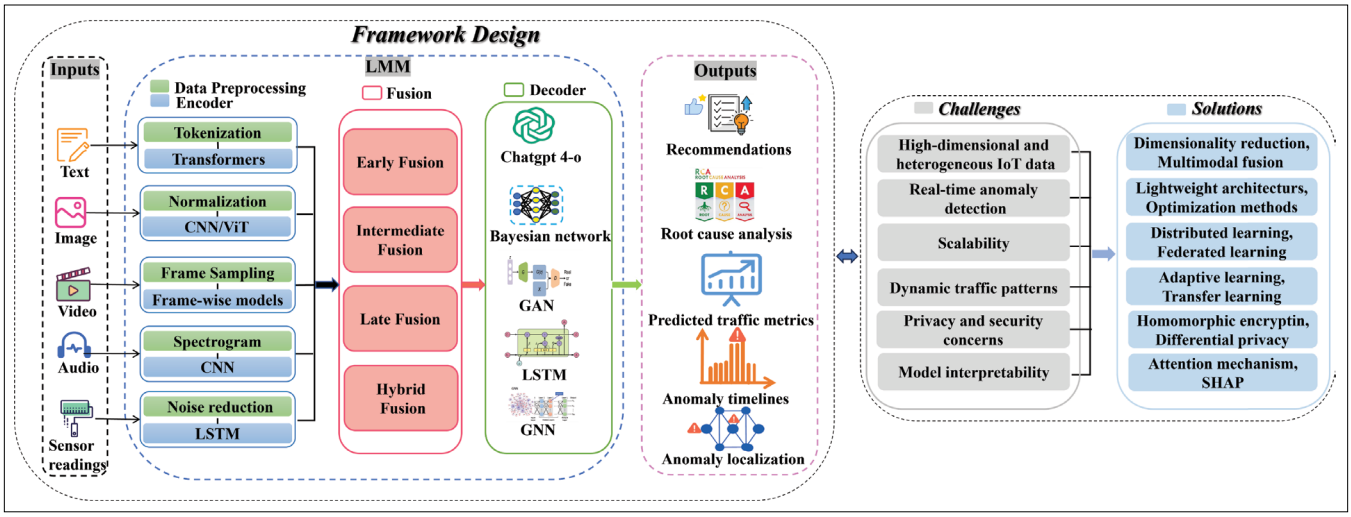


FIGURE 2. LMM-based Anomaly Detection.

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

allowing the model to exploit shared information across modalities while preserving modality-specific features. Late fusion combines the outputs of separate models trained on each modality, facilitating independent learning but still benefiting from complementary information. Hybrid fusion strategies, which combine elements from multiple fusion types, further enhance the model's ability to process complex multimodal data.

Once the data is fused, the model uses advanced machine learning techniques to perform *traffic prediction*. This involves not only predicting traffic volumes but also anticipating the flow dynamics across different IoMT devices and network segments. Techniques such as recurrent neural networks (RNNs) or long short-term memory (LSTM) networks can be used to capture temporal dependencies in traffic patterns, while convolutional neural networks (CNNs) can help extract spatial features from traffic data. Additionally, graph-based neural networks (GNNs) are increasingly applied in IoMT to model the relationships between different network devices, allowing for a more accurate representation of traffic flows within complex network topologies. The fusion of multimodal data, including location-based channel state information (CSI), environmental conditions (e.g., weather, temperature, and obstacles), and real-time traffic data, allows the model to predict traffic congestion and potential bottlenecks with high accuracy. These predictions are then compared with baseline patterns or real-time measurements to identify anomalies and trigger alarms. Detected anomalies are localized by correlating network device states with alarm transitions, enabling the identification of root causes and providing actionable insights for network management. This process enhances the reliability, efficiency, and security of IoMT systems, ensuring more responsive and optimized traffic flow in highly dynamic environments.

Challenges and Solutions

Despite its potential, the deployment of LMM-based anomaly detection in IoMT environments presents several key challenges. First, *IoT data is inherently high-dimensional and heterogeneous*, with diverse data modalities collected from various devices. This complexity makes it challenging to retain meaningful features while discarding irrelevant ones. Dimensionality reduction techniques and multimodal fusion strategies can address these issues, but achieving synchronization across temporal and spatial modalities remains an open problem. Second, *real-time anomaly detection* is critical for ensuring the responsiveness of IoMT systems. However, the resource-intensive nature of LMMs poses significant obstacles to achieving real-time performance. Lightweight architectures and optimization methods, such as pruning, quantization, and knowledge distillation, can help accelerate inference while maintaining acceptable accuracy. However, improper application of these techniques may compromise model performance.

Scalability is another critical challenge, as IoMT networks often encompass a large number of devices with heterogeneous characteristics. Distributed and federated learning approaches can enable scalable anomaly detection by leveraging edge computing and preserving data privacy. However, these methods face limitations, such as inefficient aggregation mechanisms and security vulnerabilities. *Dynamic traffic patterns* further complicate the deployment of anomaly detection systems. Adaptive learning techniques, such as online learning and reinforcement learning, enable models to evolve with changing traffic conditions. However, these methods may encounter issues with asynchronous inputs and expanded action spaces. Transfer learning offers another approach to reduce retraining time, though it may fail when source and target domains differ significantly.

Additionally, *privacy and security concerns* are paramount in IoMT environments due to the sensitive nature of IoT data. Mechanisms such as homomorphic encryption and differential privacy provide robust protections but can introduce computational overhead or degrade model utility if not properly managed. Lastly, the *interpretability of LMM-based models* is essential for building trust and facilitating operational decision-making. Explainable AI techniques, such as attention mechanisms for model-intrinsic interpretability and model-agnostic methods like Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), can enhance interpretability. However, these methods often struggle with high-dimensional or multimodal data, as SHAP and LIME are primarily designed for structured tabular or textual inputs rather than complex multimodal interactions.

In conclusion, LMM-based traffic prediction provides a robust foundation for anomaly detection in IoMT. Addressing challenges related to data heterogeneity, real-time performance, scalability, dynamic traffic patterns, privacy, and interpretability will be critical to unlocking the full potential of this approach. Continued research into adaptive learning models, efficient fusion strategies, and privacy-preserving mechanisms will further enhance its practicality and effectiveness.

LMM-Based Data Fusion for Physical Layer Security (PLS) in IoMT

Proposed Framework

This section explores the application of LMMs for multimodal data fusion in IoMT to enhance PLS. The framework addresses two primary scenarios: identifying attackers and eavesdroppers, whose locations may or may not be known, and employing traffic jamming to secure legitimate users while minimizing the effective signal received by eavesdroppers. By integrating diverse signals from near-field communication (NFC) and far-field communication (FFC) sources, such as base stations, Wi-Fi networks, cell-free cognitive architectures, intelligent reflective surfaces (IRS), and other ad hoc communication devices, the framework leverages LMMs to provide a robust and adaptive solution. It accounts for the varying characteristics of these signals, including NFC, FFC, and reflected sources, enabling LMMs to optimize security measures and enhance PLS performance across different communication ranges, as illustrated in Fig. 3.

The process begins with signal collection and preprocessing, where analog or digital signals

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

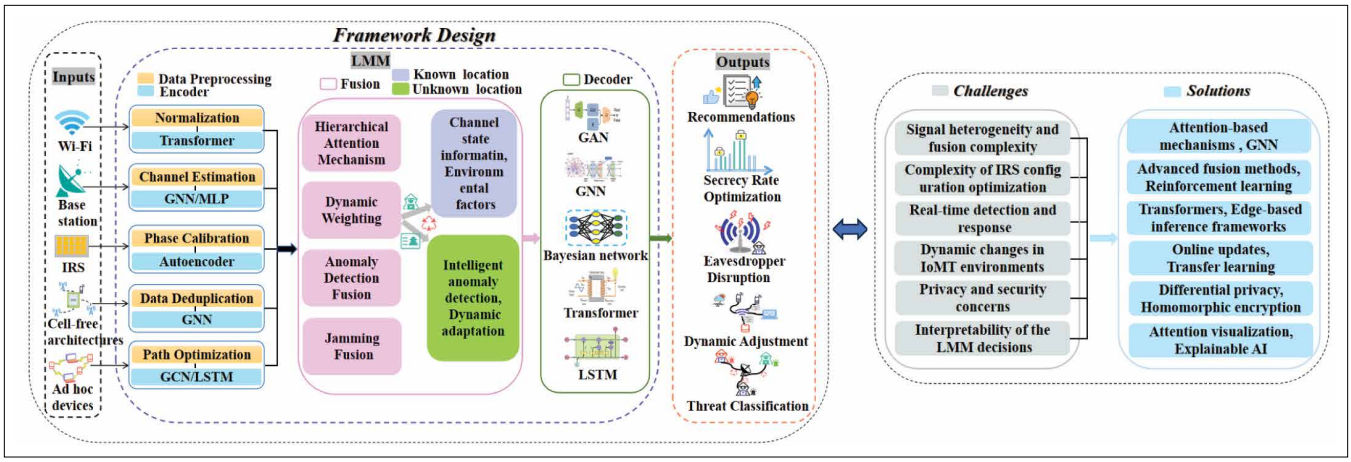


FIGURE 3. LMM-based Data Fusion for PLS in IoMT.

from various sources, such as base stations and WiFi, undergo normalization, noise reduction, and temporal alignment. Normalization here involves adjusting the amplitude or power of the raw signals so that they are within a comparable range, typically by scaling the signals to a standard reference level. This ensures that signals with different transmission power levels or measurement units can be effectively integrated, avoiding any one signal from dominating the fusion process. These preprocessing steps ensure the data is ready for integration into the fusion-enabled model, strengthening PLS robustness. Using advanced multimodal fusion techniques like hierarchical attention mechanisms, which selectively focus on important features at different levels, and dynamic weighting, which adjusts the contribution of each signal based on its quality or relevance, the framework combines complementary features across modalities. This integrated signal representation enhances anomaly detection and improves PLS by identifying potential threats, even without knowing the locations of attackers or eavesdroppers.

When the locations of attackers and eavesdroppers are known, LMMs use location information to optimize jamming and other defense strategies, enhancing PLS. By integrating multimodal data, such as location-based CSI and environmental factors (e.g., channel attenuation, weather, temperature, and obstacles), the framework adapts transmission parameters to minimize signal leakage to eavesdroppers. This spatial and environmental awareness enables targeted countermeasures, improving secrecy rates for legitimate users while minimizing network impact, providing a more efficient and reliable security solution. When their locations are unknown, LMMs enhance PLS through anomaly detection and dynamic adaptation. The framework integrates signals from sources like base stations, Wi-Fi networks, and IRS to detect malicious activity. By fusing NFC and FFC data, LMMs can infer potential attacks without location information. Environmental factors, such as multipath propagation, weather, and interference, further improve threat detection.

Challenges and Solutions

The implementation of LMM-based data fusion for PLS in IoMT presents several intertwined challenges that require innovative solutions. First, *the heterogeneity of signals from diverse sources*, such as base stations, Wi-Fi networks, cell-free systems, NFC, FFC, and IRS, complicates the task of maintaining critical features during integration. NFC and FFC signals differ significantly in their propagation characteristics—NFC typically operates over short distances, while FFC covers long distances—resulting in varying degrees of signal attenuation and interference. Integrating these signals requires sophisticated fusion techniques, like attention-based mechanisms or graph neural networks (GNN), which must balance computational efficiency with accuracy. Additionally, *optimizing IRS configurations* presents significant challenges. Achieving precise phase control for IRS elements to enhance signal quality and security must account for factors like multipath propagation and interference. The high element count and real-time configuration demands further complicate this task. Methods such as advanced fusion and reinforcement learning offer potential solutions but require careful design to ensure efficiency and adaptability in diverse signal environments.

Real-time detection and response remain key challenges due to the high computational demands of LMMs. Detecting potential attackers or eavesdroppers in real time, especially when their locations are unknown, requires responsive models that can quickly adapt to changing environments. Lightweight architectures, such as distilled transformers or edge-based inference frameworks, can improve responsiveness. However, optimizing these methods to balance accuracy and reduce latency is a significant challenge.

Dynamic changes in IoMT environments complicate LMM deployment. Environmental factors such as weather, interference, and the movement of obstacles (e.g., people or vehicles) can rapidly alter signal patterns. Adaptive learning techniques, including online updates and transfer learning, allow the model to evolve with these fluctuations. While offering flexibility, ensuring stability during rapid changes and minimizing retraining overhead are key challenges. In particular, *the selection and placement of IRS elements* need to be optimized based on real-time environmental feedback. The framework should dynamically select the most effective IRS elements to maintain optimal performance in changing environments.

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

Privacy and security concerns are paramount, especially in sensitive domains like healthcare. Techniques such as differential privacy and homomorphic encryption safeguard data, but integrating them into LMM-based frameworks introduces computational overhead. Effective implementation requires balancing privacy with performance. For example, real-time jamming and countermeasures against eavesdroppers should ensure that the privacy of legitimate users is not compromised.

Lastly, *Interpretability of LMM decisions* is crucial for fostering trust and operational reliability. Techniques like attention visualization and explainable AI methods can enhance understanding of the model's outputs, but they need to evolve to handle the complexities of multimodal data fusion. Explaining why certain IRS configurations or security measures are applied is essential for ensuring user confidence, especially in critical applications like healthcare.

Case Study

In the previous sections, we discussed the use of LMMs for anomaly detection and PLS enhancement in IoMT networks. We now present a case study demonstrating anomaly detection in IoMT environments.

Proposed Anomaly-Alarm Correlation-Based Anomaly Detection (AAC-AD) Framework

Anomalies can affect multiple boards (hardware units or device nodes), disrupting traffic flows and triggering widespread alarms across interconnected boards. These alarms may originate from local anomaly detection or propagate from remote boards, significantly increasing the complexity of anomaly detection in IoMT networks due to the large volume of alarms and their interdependencies.

Fig. 4(1) illustrates anomaly-alarm propagation. The anomalous link A_1 is detected by board D, triggering alarms a_4 and a_1 upstream and downstream. The anomaly then propagates to boards F and G, triggering alarms a_3 and a_2, respectively. Fig. 4(2) shows another example. The minor anomaly A_2 on board B does not trigger alarms but notifies neighboring boards C, D, and E, causing alarms a_4, a_3, and a_2. Board F generates alarm a_1 due to the propagation of A_2. Despite affecting the same topology, A_1 and A_2 create identical alarm patterns due to different traffic distributions. The root alarm and its relationships with other alarms depend on the anomaly type and network state. For instance, in Fig. 4(1), a_1 triggers a_3, while in Fig. 4(2), a_3 triggers a_1.

The alarm propagation process is modeled using the anomaly-alarm correlation tree (AACT) shown in Fig. 4, which identifies dependent alarms and improves anomaly detection performance. In this study, a *board instance* is characterized by its board ID and type, while an *alarm instance* is defined by its occurrence time, type, board ID, and board type. Our focus is on assessing correlations between instances, including *board* → *alarm* (anomalous board triggers an alarm) and *alarm* → *alarm* (one alarm triggers another).

Fig. 5 illustrates the AAC-AD flowchart. Features are extracted from each instance pair in the historical dataset (Fig. 5(i)), with the extraction method detailed in the section "Feature Extraction." The processed dataset is then used to train a binary classifier that captures the correlation for each pair (Fig. 5(ii), discussed in the section "Binary Classifier.". Finally, correlations are identified through the AACT formulation (Fig. 5(iii), detailed in the section "AACT Formulation"), producing one or more AACTs as the output.

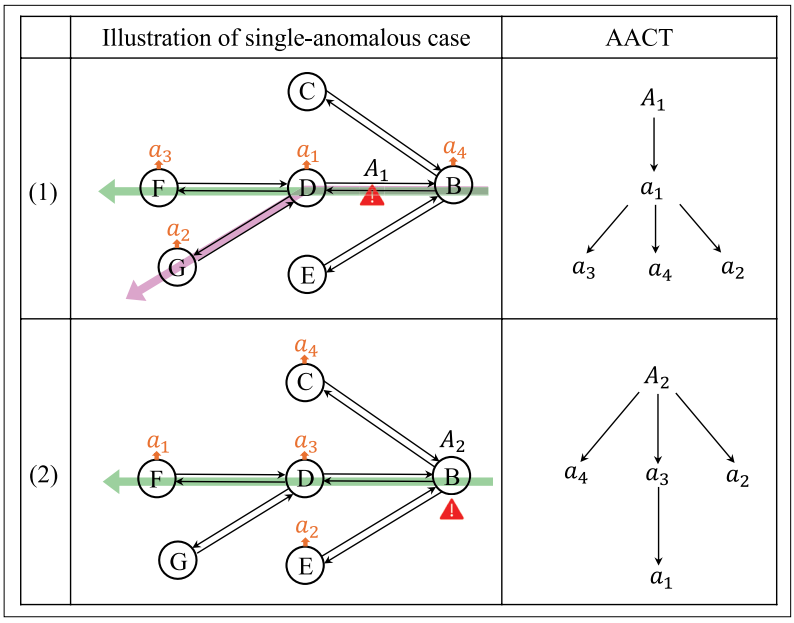


FIGURE 4. Two cases of single-board anomaly are presented, where A, …, F represent boards, purple and green arrows indicate traffic flows, a_1, a_2, a_3, and a_4 denote alarm types, and the caution symbol labels the anomalous link or board.

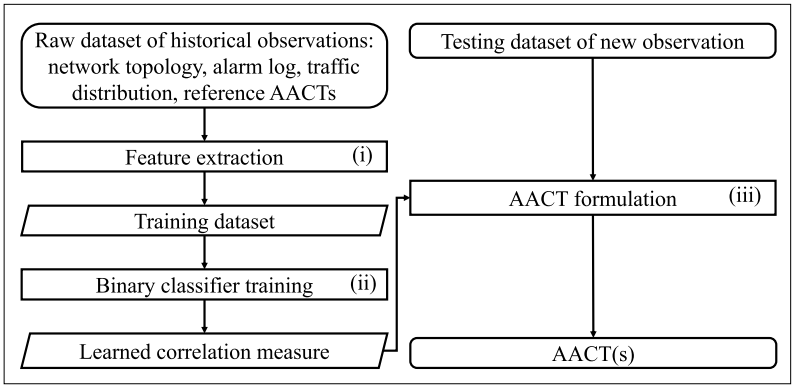


FIGURE 5. Flowchart of AAC-AD.

1) Feature Extraction: The raw dataset consists of historical observations from each observation window (OW), linked to a specific network topology, alarm log, traffic distribution, and reference AACTs. The network topology defines the physical structure at the board level, including all boards and interconnections. The alarm log records all alarm instances, while traffic distribution represents the flow of traffic between source and destination boards. Instance pairs are formed by combining board and alarm instances, and the ground-truth correlation set for training is derived from observed correlations in the reference AACTs.

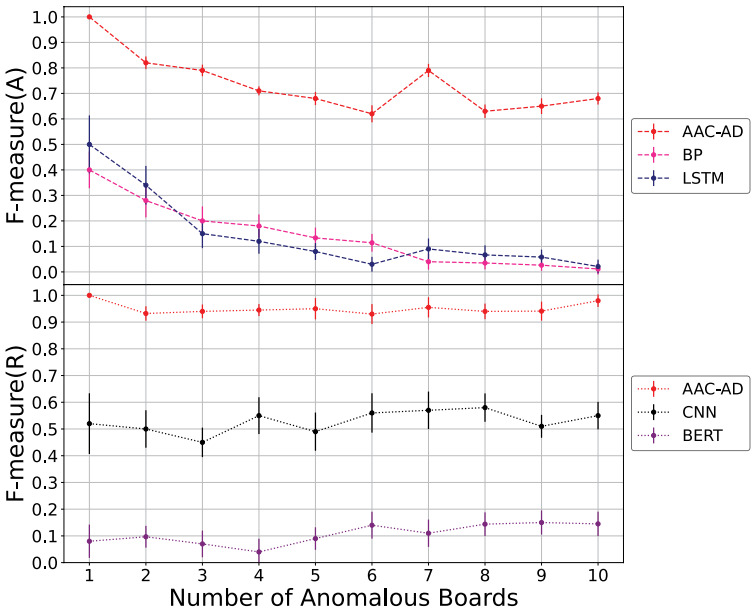
A multimodal feature space, which combines temporal and spatial information, is created to

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

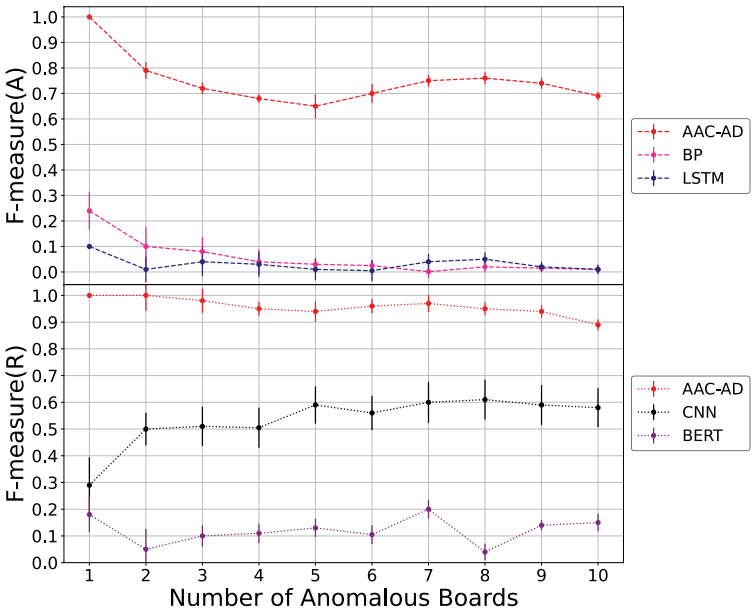
describe each instance pair and serves as input to the binary classifier. The key features of the space include *propagation delay*, representing the temporal interval between two instances; *topological distance*, indicating the shortest physical path between the boards of the two instances; *traffic flow label*, a binary value indicating whether the boards share a common traffic flow; and *instance attributes*, reflecting the characteristics of the board and alarm instances forming the pair. By mapping instance pairs within each OW to this feature space, transformed datasets are generated. These datasets, collected over multiple consecutive OWs, are then aggregated to construct the training dataset for the binary classifier.

Method	# of trainable parameters	F-measure(R)	F-measure(A)
AAC-AD	5435	1.000±0.000	1.000±0.000
CNN	327541	0.734±0.083	N/A
BERT	2437810	0.132±0.142	
BP	43428	N/A	0.824±0.182
LSTM	68568		0.882±0.137
GNN	50239		0.342±0.254

(a)



(b)



(c)

FIGURE 6. a) Result on the benchmark dataset. b) Result on the regional anomaly dataset under various traffic distributions. c) Result on the regional anomaly dataset under various network topologies.

2) Binary Classifier: The model uses a 3-layer fully connected neural network (FCNN) with hidden layers of 32, 32, and 64 units. An embedding layer maps categorical features into a continuous space, which is then combined with numerical features and passed through the FCNN. The classifier outputs a probability between 0 and 1, with higher values indicating a stronger likelihood of correlation between the instances.

3) AACT Formulation: For each OW, the testing dataset, which includes network topology, alarm logs, and traffic distribution, is used to construct one or more AACTs to maximize alarm instance coverage. Each AACT is rooted at a board instance and spans multiple alarm instances as leaves.

Using the testing dataset and pre-trained binary classifier, a weighted directed acyclic graph (WDAG) is constructed as the search space. Its vertices represent instances, and the arcs depict potential correlations of type *board* → *alarm* or *alarm* → *alarm*. Arc weights, which range from 0 to 1, reflect correlation costs. These are calculated either as the complement of the classifier-predicted probability for *alarm* → *alarm* correlations or the intersection probability with the board's time-to-anomaly density function for *board* → *alarm* correlations.

The WDAG is pruned by removing arcs with weights above 0.5 and retaining only the minimum-weight *alarm* → *alarm* arc for each alarm instance. This results in a reduced WDAG where the alarm instances form a forest. Root alarm instances are identified as those connected only to board instances, and their associated board sets are merged to form a refined board instance set. A bipartite graph between the refined board instance set and root alarm instances is processed to form a board-root alarm instance forest. This step minimizes the number of board instances needed to cover all root alarms. Selected board instances serve as roots, while root alarm instances form the leaves. Combining this forest with the alarm instance forest generates the final AACT(s).

To ensure comprehensive detection, the union of all possible AACT solutions is taken as the output. Boards without correlated alarms are excluded from being marked as anomalous. Due to the nature of alarm propagation, only *One-2One* or *One2Many* correlations are allowed, while *Many2One* relationships are prohibited.

Simulation

This study evaluates the generality and transferability of AAC-AD by testing its binary classifier, trained on a default network state, across various scenarios with differing network topologies, traffic distributions, and anomaly conditions. For

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions

comparison, alternative methods such as back-propagation neural networks (BP), long short-term memory networks (LSTM), convolutional neural networks (CNN), graph neural networks (GNN), and bidirectional encoder representations from transformers (BERT) are considered.

Ground-truth alarms and reference AACTs are generated using a simulator, which models anomalies based on predefined propagation rules, network topology, and traffic distribution within an *OW*. The simulator assumes each anomaly independently affects one or more boards, disrupting traversing traffic flows.

1) Data Preparation: The raw dataset is generated based on a default network state comprising 7 nodes, 320 boards, and 30 traffic flows. A total of 2840 alarms are generated by independently inducing anomalies in each board within the network topology. To validate the adaptability of AAC-AD, three testing datasets are generated. A benchmark dataset of 10 single-board anomaly from the default network state is used to evaluate the baseline performance. Two regional anomaly datasets are then produced from 20 new network states: 10 states result from varying the number of traffic flows from 10 to 100 in a consistent network topology, while the remaining 10 states are derived by fixing the number of traffic flows at 50 and altering the number of boards from 304 to 600. In each state, a regional anomaly is generated by increasing the number of anomalous boards to 10. Boards are selected sequentially, starting with a random board and expanding to connected boards one at a time. The performance is evaluated using the harmonic mean of precision and recall, for two key outcomes: *F-measure (R)*, which measures the detection accuracy of root alarms, and *F-measure (A)*, which assesses the identification of anomalous boards.

2) Results: As shown in Fig. 6(a), AAC-AD outperforms the counterparts with minimal trainable parameters. Moreover, Fig. 6(b) and (c) show that AAC-AD performs best in identifying root alarms and anomalous boards. CNN and BERT falter due to neglecting traffic distribution and spatial relations, while BP and LSTM degrade as anomalous boards increase, ignoring alarm attributes. Unlike counterparts requiring retraining for new network states, AAC-AD's classifier, trained once, achieves the best migratability.

Conclusion

In this paper, we demonstrated how LMMs can effectively address critical challenges in securing IoMT networks by enhancing anomaly detection and PLS. Through multimodal data fusion, LMMs can integrate diverse data sources, improving the accuracy of traffic prediction and enabling real-time detection of anomalies. This approach facilitates early identification of cyber threats and system failures, ensuring robust and adaptive responses to security challenges. Additionally, we explored how LMMs optimize PLS by identifying vulnerabilities, dynamically allocating resources, and strengthening network resilience against eavesdroppers and adversaries. Despite the promising results, challenges such as data heterogeneity, computational efficiency, privacy concerns, and the interpretability of LMM-based decisions remain significant hurdles. Addressing these challenges through advanced techniques like lightweight architectures, adaptive learning, and privacy-preserving mechanisms will be essential for fully realizing the potential of LMMs in IoMT.

Additionally, we explored how LMMs optimize PLS by identifying vulnerabilities, dynamically allocating resources, and strengthening network resilience against eavesdroppers and adversaries.

References

[1] S. S. Hameed et al., "A systematic review of security and privacy issues in the Internet of Medical Things; the role of machine learning approaches," *PeerJ Comput. Sci.*, vol. 7, p. e414, Mar. 2021.

[2] Y. Zhang et al., "Diversity-driven proactive caching for mobile networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 7, pp. 7878–7894, Jul. 2024.

[3] C. Hu et al., "Self-refined generative foundation models for wireless traffic prediction," 2024, *arXiv:2408.10390*.

[4] W. Jiang et al., "Mobile traffic prediction in consumer applications: A multimodal deep learning approach," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3425–3435, Feb. 2024.

[5] H. Ma and K. Yang, "MetaSTNet: Multimodal meta-learning for cellular traffic conformal prediction," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 2, pp. 1999–2011, Mar. 2024.

[6] Y. Pang et al., "Large language model-based optical network log analysis using LLaMA2 with instruction tuning," *J. Opt. Commun. Netw.*, vol. 16, no. 11, pp. 1116–1132, 2024.

[7] Y. Zhou et al., "Leveraging large language models and BERT for log parsing and anomaly detection," *Mathematics*, vol. 12, no. 17, p. 2758, Sep. 2024.

[8] Y. Chen et al., "Automatic root cause analysis via large language models for cloud incidents," in *Proc. 19th Eur. Conf. Comput. Syst.*, Apr. 2024, pp. 674–688.

[9] W. Lee and J. Park, "LLM-empowered resource allocation in wireless communications systems," 2024, *arXiv:2408.02944*.

[10] J. Tong et al., "WirelessAgent: Large language model agents for intelligent wireless networks," 2024, *arXiv:2409.07964*.

[11] J. Su, S. Nair, and L. Popokh, "Leveraging large language models for VNF resource forecasting," in *Proc. IEEE 10th Int. Conf. Netw. Softwarization (NetSoft)*, Jun. 2024, pp. 258–262.

Biographies

Yan Jiao (yanyanjiao2018@gmail.com) received the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Waterloo, Canada, in 2020 and 2024, respectively. She is currently a Post-Doctoral Researcher with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China. Her research interests focus on network fault management and machine learning.

Mian Muaz Razaq (mianmuaz97@gmail.com) received the B.S. degree in software engineering from COMSATS University Islamabad, Pakistan, in 2019, and the master's degree in computer science from Kyungpook National University, South Korea, in 2021, where he is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering. His research interests include edge computing, B5G communications, and physical layer security.

Limei Peng (auroraplm@gmail.com) is currently a Full Professor with the School of Computer Science and Engineering, Kyungpook National University (KNU), Daegu, Republic of Korea. Her research interests include cloud computing, fog computing, data center networks, the Internet of Things (IoT)/the Internet of Vehicles (IoV), and 5G communication networks.

Pin-Han Ho (Fellow, IEEE) (pinhanho71@gmail.com) is currently a Full Professor with the Department of Electrical and Computer Engineering, University of Waterloo. He is the author/co-author of over 400 refereed technical articles, several book chapters, and the co-author of two books on internet and optical network survivability. His current research interests cover a wide range of topics in broadband wired and wireless communication networks, including wireless transmission techniques, mobile system design and optimization, and network dimensioning and resource allocation. He is a Professional Engineer Ontario (PEO).

ized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on February 18,2026 at 10:21:21 UTC from IEEE Xplore. Restrictions